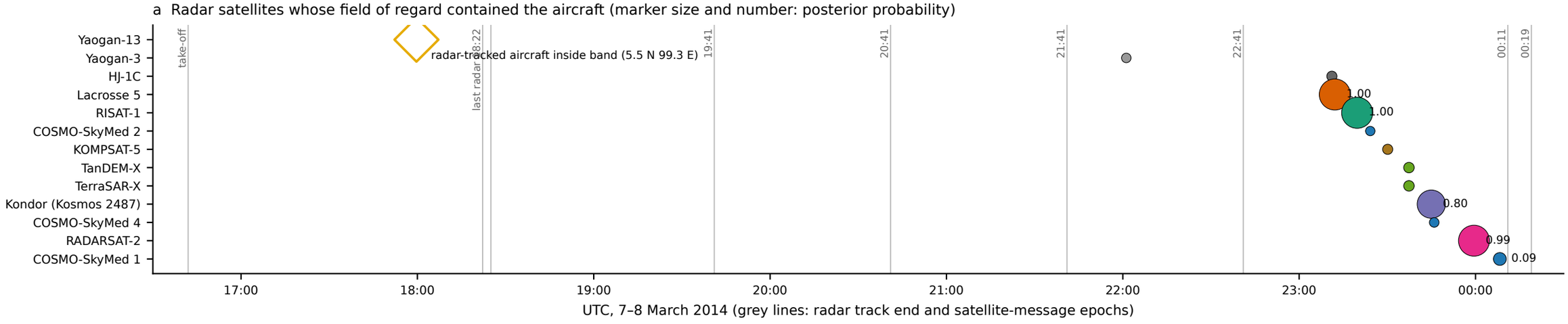
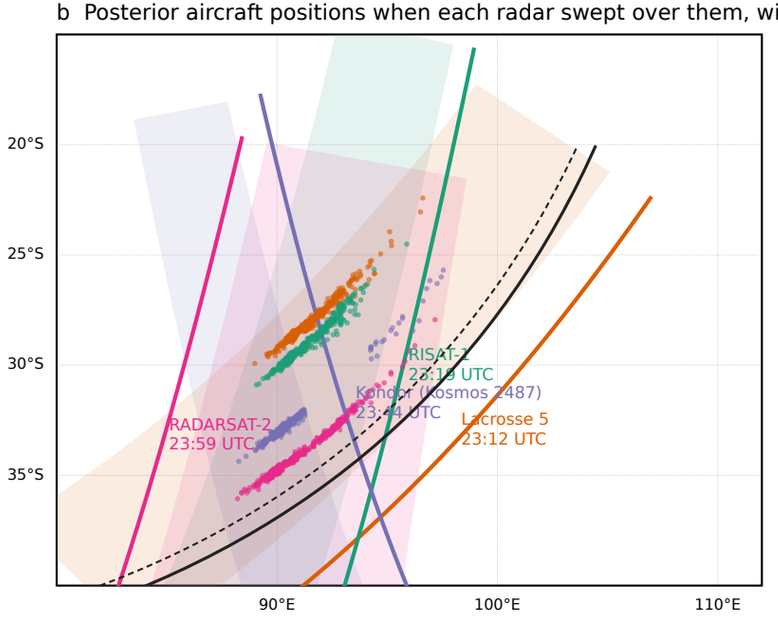
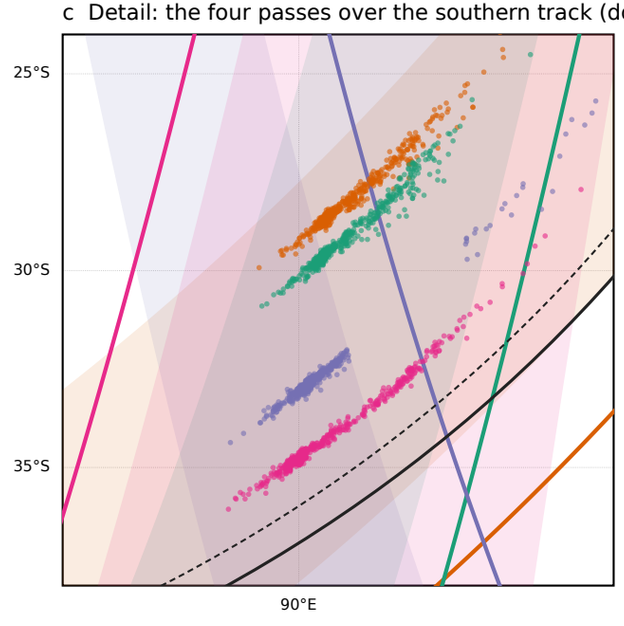
Radar satellites that could have imaged MH370 in flight, 7–8 March 2014

A radar is counted when the line straight across its track swept over the aircraft's position while that position lay inside the radar's range of incidence angles on a side it can look to. Black solid: 7th arc; dashed: 6th arc.







Plain-language note: four radar satellites passed while the aircraft was almost certainly inside the area they could image: Lacrosse 5 (USA) at 23:12, RISAT-1 (India) at 23:20, Kondor, Kosmos 2487 (Russia) at 23:45 and RADARSAT-2 (Canada) at 23:59 UTC. Between 18:25 and 22:00 no radar had the aircraft in reach. Earlier, at 17:59, a Chinese Yaogan-13 radar had the radar-tracked aircraft near Penang inside its reach, which makes that pass a possible check of whether such images show an airliner. This is what could have been imaged, not what was imaged: a radar records only when ordered to, and each image covers one strip inside the area shown.

Technical: aircraft positions from core run J routes.npy (2,000 posterior routes per seed, 10-min vertices 18:01:49–00:11:49, linear interpolation; waypoints family 48,000 routes, P(family) 0.9982; probabilities are the family-weighted mixture) and the Geoscience Australia radar track (16:42:27–18:22:12). Satellite positions: SGP4, 5 s steps; zero-Doppler crossing found per route by sign change of the along-track coordinate. Bands: SAR incidence ranges from operator specifications (RADARSAT-2 10–60°, RISAT-1 12–55°, grade B) or open-literature estimates (Lacrosse 5 20–60°, Kosmos 2487 20–55°, both sides, grade C). Route chords cut turns (about 25 NM tolerance); shares within 25 km of a band edge: Lacrosse 0, RISAT-1 0, RADARSAT-2 0.001, Kosmos 2487 0.03. Altitude not stored in routes; FL350 assumed for imaging geometry.